Tô Gia Bảo

Data Engineer | AML Analyst

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Summary

Data Engineer in digital banking with hands-on AML and fraud analytics experience. Builds reliable ETL pipelines, data lakes, and dimensional models using Python, SQL, and AWS, alongside transaction monitoring and behavioral risk scoring. Reduced S3 table storage by 45% and false-positive triage workload for Compliance and Fraud Operations by approximately 50%.

Experience

Timo Digital Bank by BVBank — *Ho Chi Minh City, Vietnam*

Data Engineer — *Mar. 2025 – Present*

- Implemented MariaDB data pipelines spanning ingestion with Amazon EMR and AWS Glue, PII masking in the curated layer, and downstream dbt transformations, turning source data into analytics-ready datasets for reporting and analysis.
- Optimized an S3 table with Parquet ZSTD-3 and compaction, saving 38.4 TB (45%) and cutting object count by 80% across 56.8 billion rows (2023 to Sep. 2026).
- Built a daily incremental AML and fraud feature mart on Athena and dbt, tracking 40+ behavioral features across millions of transactions to detect fund pass-through, structuring, and unusual arrival patterns.
- Developed a 10-signal mule-account scoring engine covering transaction velocity, nighttime bursts, small incoming transfers, and concentrated cash-outs. Added balance retention gates and two-tier review queues, cutting false-positive triage workload for Compliance and Fraud Operations by approximately 50% while retaining confirmed mule cases for review.
- Built daily alerts comparing intraday and month-to-date transaction frequency and gross value against a three-month baseline to flag behavioral spikes.
- Modeled dbt facts and dimensions with defined grains, keys, and relationships for Lightdash dashboards covering Airflow reliability, SLAs, Athena usage, and S3 storage.
- Validated transformations and integrations with dbt tests, Great Expectations, and pytest, checking query contracts, SQL compilation, and dbt/Lightdash metadata consistency.
- Built a FastMCP server for self-service Slack analytics using dbt docs, Lightdash, and semantic metadata.
- Automated MongoDB backup validation and general ledger lookups with Python and Airflow, reducing manual checks.
- Migrated Timo Metrics from Loki/Grafana to SigNoz for centralized logs, metrics, and troubleshooting.
- Delivered Tracking247 to sync Customer-provided shipment IDs and status updates into MongoDB via Airflow and Streamlit.

Projects

Mini Lakehouse — *GitHub*

Lakehouse & Platform Engineering — *Jul. 2026 – Present*

- Built a contract-driven AWS lakehouse on S3, Iceberg, Glue, Athena, and EMR Serverless, with Airflow and platform services self-hosted on OCI and infrastructure managed through Terraform.
- Delivered batch and streaming pipelines for GitHub activity, arXiv research, and SSI market data, with dbt transformations, Lightdash and Sigma BI dashboards, Modal GPU OCR, and Netdata monitoring via Cloudflare Zero Trust.

VN Market Pulse — *GitHub*

AI Research & Content Platform — *Jul. 2026*

- Built a Vietnamese research pipeline with multi-source search, deduplication, and reranking to generate source-backed posts.
- Used PydanticAI for typed output and validation, with retrieval budgets, retries, usage tracking, and CLI/Streamlit interfaces.

Technical Skills

Programming & Processing	Python, SQL, Spark/PySpark, Databricks, Amazon EMR
Storage & Modeling	Amazon S3, Apache Iceberg, PostgreSQL, MariaDB, MongoDB; dimensional modeling
Pipelines & Analytics	Airflow, dbt, AWS Glue, Azure Data Factory (ADF), Amazon Athena, Lightdash
Data Quality & Testing	dbt tests, Great Expectations, pytest
DevOps & Observability	Terraform, Docker, Git, SigNoz, Netdata
AML & Fraud Analytics	Transaction monitoring, mule account detection, behavioral risk scoring, alert prioritization

Education

University of Science, VNU-HCM (HCMUS) — *Ho Chi Minh City, Vietnam*

Bachelor of Data Science | GPA: 3.78/4.00 — *2022 – 2026*

Awards & Certifications

2025 **Top 25**, AIC Challenge
2025 **Recipient**, Ninety Eight Scholarship
2024 **Top 10**, RMIT Data Analytics Competition
2024 **Google Business Intelligence**, Professional Certificate
2024 **Google Data Analytics**, Professional Certificate

Publication

GalaxyAssistant: An Intelligent Assistant for Multimedia Event Retrieval — *SOICT*

Co-author — *2025*

Languages

English TOEIC Listening & Reading: 910 | TOEIC Speaking & Writing: 350